

24ADC001 — BUSINESS INTELLIGENCE

LAB ASSIGNMENT_21.08.2026

One scenario, one dataset, one tool each: Power BI, Tableau, OpenAI, and NotebookLM

LAB 1 — POWER BI

PedalShare: Building an OLAP Cube for a City Bike-Share Network

Concepts covered: data warehouse star schema, OLAP (roll-up, drill-down, slice, dice, pivot), analytic functions (DAX).

Scenario

PedalShare runs a bike-share network across Coimbatore, Chennai, and Bengaluru. Their ops team currently gets a single flat spreadsheet dump every morning and manually eyeballs it — there's no model, no drill-down, no way to compare cities or rider types quickly. You've been brought in to build them their first real analytical model: a proper star schema in Power BI that lets any manager slice the data six different ways before their coffee gets cold.

Dataset: pedalshare_trips.csv

- trip_id, date, city, station_start, rider_type (Member/Casual), bike_type (Classic/Electric), duration_min, weather, temperature_c
- Roughly 1,500 trips across 3 cities over a 90-day period, with real embedded patterns: casual riders take longer trips on weekends, rain shortens trips, and Bengaluru runs a higher volume than the other two cities.

Tasks

1. Import the CSV and use Power Query to split it into a Fact_Trips table and at least two dimension tables (e.g., Dim_Date, Dim_City, Dim_RiderType).
2. Build the relationships in Model view to complete the star schema.
3. Build a Matrix visual with City on rows and Month/Week on columns, and use it to roll up to monthly totals and drill down to weekly, then daily detail.
4. Add slicers for rider_type and weather. Produce one sliced view (a single weather condition) and one diced view (a specific city + rider type + weather combination).
5. Write a DAX measure using RANKX() to rank stations by total trips, and a second measure computing average trip duration by rider type.
6. Investigate: does casual ridership really spike on weekends across all three cities, or is that pattern driven by just one city? Show the visual that answers this.

Deliverables

- The .pbix file with the completed model, matrix visual, slicers, and both DAX measures.
- A short write-up answering the weekend-ridership investigation question, with a screenshot of the supporting visual.

LAB 2 — TABLEAU

CityCare Hospital: A Decision Support Dashboard for the Emergency Department

Concepts covered: business intelligence dashboards, decision support systems, types of decisions (structured/semi-structured/unstructured).

Scenario

CityCare Hospital's Emergency Department is chronically short-staffed at the worst possible times, and nobody can say with confidence when those times actually are. The department head has a hunch it's "weekend nights," but hunches don't get budget approved. You've been asked to build a dashboard that turns the hunch into evidence — and to recommend, using it, what kind of decision-making process the hospital should actually use to solve the problem.

Dataset: citycare_er_visits.csv

- visit_id, date, arrival_hour, department, triage_level (1=most critical to 5=least), wait_time_min, age_group, outcome, day_of_week
- Roughly 1,700 visits over 120 days, with embedded patterns: wait times scale with triage severity, and weekend nights (6pm–11pm) run heavier volume with a wait-time surge.

Tasks

1. Connect Tableau to the CSV and build three linked worksheets: a heatmap of visit volume by day-of-week × hour, a bar chart of average wait time by department, and a breakdown of outcomes by triage level.
2. Build a drill-down hierarchy on Department → Triage Level, and demonstrate expanding and collapsing it.
3. Add a filter for day-of-week and a parameter to toggle between viewing wait time vs. visit volume — use these to isolate the weekend-night pattern specifically.
4. Create a calculated field ranking departments by average wait time using a table calculation.
5. Combine the views into one dashboard with an action filter connecting them.
6. Write a 200-word recommendation: is "when to add ER staff" a structured, semi-structured, or unstructured decision, based on what the dashboard actually shows? What role should a DSS play here versus a manager's judgment?

Deliverables

- The Tableau packaged workbook (.twbx).
- The 200-word written recommendation.

LAB 3 — OPENAI (CHATGPT)

SupportDesk: Turning Trapped Tacit Knowledge Into a Rule-Based Troubleshooting Guide

Concepts covered: knowledge management (tacit vs. explicit, the SECI model), rule-based models.

Scenario

SupportDesk's customer support agents have quietly built up years of troubleshooting know-how, but almost none of it is written down anywhere official — it lives in scattered ticket notes and in each agent's head. New hires take months to become as fast as the veterans, simply because nobody

has ever externalized what the veterans already know. You've been asked to use an AI tool to read a sample of resolved tickets and draft the beginnings of an official, rule-based troubleshooting guide — exactly the externalization step the SECI model describes.

Dataset: supportdesk_tickets.csv

- ticket_id, issue_category, agent_notes (free-text resolution notes), resolution_time_min, resolved_first_contact (Y/N), customer_satisfaction (1–5)
- 135 real-sounding resolved tickets across five categories: Login Issue, Billing, Bug Report, Feature Request, and Performance.

Tasks

1. Upload or paste the agent_notes for the Login Issue and Billing tickets into ChatGPT. Ask it to identify the 3–5 most common root causes described across the notes.
2. For each root cause identified, prompt ChatGPT to draft an IF-THEN rule in the style covered in class (e.g., “IF customer reports X AND Y THEN try Z first”) that a new agent could follow without needing the tacit experience the notes came from.
3. Cross-check: pick 5 tickets ChatGPT didn't explicitly reference, and test whether your drafted rules would have actually resolved them. Note any rule that fails and refine it.
4. Ask ChatGPT to flag which tickets look like they reveal a product bug rather than a one-off user error — compare its picks against the Bug Report category and see how well they line up.
5. Write a short reflection: which parts of the agents' knowledge were easy to turn into clean rules, and which parts resisted being reduced to a rule at all? Why might that be?

Deliverables

- The drafted rule set (5–10 IF-THEN rules) with the ticket evidence behind each one.
- The cross-check results and the reflection summary (150–200 words).

LAB 4 — NOTEBOOKLM

Which Forecasting Method Actually Wins, Region by Region?

Concepts covered: business forecasting model comparison, logic-driven vs. data-driven reasoning, synthesis and interpretation.

Scenario

A national retailer tested six different forecasting methods against five years of regional sales data and logged the accuracy of each. Nobody has ever sat down and actually synthesized the results into a recommendation — the accuracy numbers just sit in a spreadsheet, and the regional analysts' notes about what makes each region's demand pattern unusual sit in a completely separate document. Your job is to bring an AI notebook tool to bear on both sources at once and produce the executive summary nobody's written yet.

Datasets: forecast_method_accuracy.csv and region_notes.csv

- `forecast_method_accuracy.csv`: region, method, mape_percent, rmse, method_description — all 5 regions tested against all 6 forecasting methods covered in Unit III (30 rows).
- `region_notes.csv`: a short analyst note per region describing its demand pattern in plain language (e.g., strong festive-season spikes, flat demand, monsoon-season surge).

Tasks

1. Upload both CSVs to a new NotebookLM notebook as sources (you may also paste in the raw text if direct CSV upload isn't available in your version).
2. Ask NotebookLM: “Which forecasting method performs best in each region, and does that match what the analyst notes say about each region's demand pattern?”
3. For any region where the best-performing method surprises you, ask NotebookLM to explain, using only the provided sources, why that method might be winning there.
4. Ask NotebookLM to identify the one region where the simplest methods (naive or simple exponential smoothing) actually perform competitively — and to explain, from the analyst note, why added complexity isn't paying off there.
5. Generate an audio overview (or a written executive summary if audio isn't available) recommending one default forecasting method per region for the retailer to adopt.
6. Write a 150-word critique: where did NotebookLM's synthesis genuinely add insight beyond just reading the spreadsheet yourself, and where did it just restate the numbers?

Deliverables

- The notebook's region-by-region method recommendations (exported or copied into a document).
- The audio overview link and written executive summary, plus the 150-word critique.

Dataset Files Provided

File	Contents
<code>pedalshare_trips.csv</code>	Bike-share trip log for Lab 1 (Power BI)
<code>citycare_er_visits.csv</code>	Hospital ER visit log for Lab 2 (Tableau)
<code>supportdesk_tickets.csv</code>	Customer support ticket log with agent notes for Lab 3 (OpenAI)
<code>forecast_method_accuracy.csv</code>	Forecast method accuracy by region for Lab 4 (NotebookLM)
<code>region_notes.csv</code>	Analyst commentary per region, paired with Lab 4